GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
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WEEK 3 – DATA QUALITY ASSESSMENT

During the third week, our team focused on evaluating the quality and reliability of the collected flight dataset. Since the dataset contained millions of records, it was important to identify potential issues before performing detailed statistical analysis.

We examined the dataset for missing values, duplicate records, inconsistent entries, incorrect data types, and unusual numerical values. These issues could affect the accuracy of the analysis and lead to misleading conclusions if they were not addressed appropriately.

Using Python and Pandas, we performed data quality checks through functions such as `isnull()`, `sum()`, `duplicated()`, `dtypes`, and `value_counts()`. These

functions helped us identify missing information, duplicate observations, and inconsistencies in the dataset.

Our team also examined important variables such as departure delay and arrival delay to identify unusual values. We noticed that some flights had extremely large delay values, which required further investigation to determine whether they were genuine observations or data errors.

We discussed the difference between missing data, invalid data, and extreme but legitimate values. We understood that not every unusual observation should automatically be removed because genuine extreme delays may provide valuable information about flight operations.

The team collaborated to document the identified data quality issues and discussed appropriate methods for handling them during the cleaning stage.

By the end of the week, we had completed the initial quality assessment and gained a better understanding of the limitations and challenges of the dataset. This helped us prepare for the data cleaning and preprocessing phase.